Started on	Wednesday, 16 October 2024, 9:41 PM
State	Finished
Completed on	Sunday, 27 October 2024, 4:50 PM
Time taken	10 days 19 hours
Marks	2.00/2.00
Grade	**100.00** out of 100.00
Name	DHIRU BHAI 730423104017

Question 1

Correct

Mark 1.00 out of 1.00

Create an interface called Remote with methods switchOn(), switchOff(), play(), increaseVolume(), decreaseVolume() and create two class called TVRemote and AudioPlayerRemote which implements the interface Remote. Define the methods in their respective classes.

Give the choice for the user to select the type of remote and call the functions.

For example:

Input	Result
1	TV Switched On TV Volume Increased TV Volume Decreased Tv Switched Off
2	Audio Player Switched On Audio Player Playing Audio Player Volume Increased Audio Player Volume Decreased Audio Player Switched Off

Answer: (penalty regime: 0 %)

```
import java.util.Scanner;

interface Remote {
    void switchOn();
    void switchOff();
    void play();
    void increaseVolume();
    void decreaseVolume();
}

class TVRemote implements Remote {
    public void switchOn() {
        System.out.println("TV Switched On");
    }
    public void switchOff() {
        System.out.println("Tv Switched Off");
    }
    public void play() {
        System.out.println("TV Playing");
    }
    public void increaseVolume() {
        System.out.println("TV Volume Increased");
    }
    public void decreaseVolume() {
        System.out.println("TV Volume Decreased");
    }
}

class AudioPlayerRemote implements Remote {
    public void switchOn() {
        System.out.println("Audio Player Switched On");
    }
    public void switchOff() {
        System.out.println("Audio Player Switched Off");
    }
    public void play() {
        System.out.println("Audio Player Playing");
    }
    public void increaseVolume() {
        System.out.println("Audio Player Volume Increased");
    }
    public void decreaseVolume() {
        System.out.println("Audio Player Volume Decreased");
```

```
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int choice = scanner.nextInt();

        Remote remote;
        if (choice == 1) {
            remote = new TVRemote();
            remote.switchOn();
            remote.increaseVolume();
            remote.decreaseVolume();
            remote.switchOff();
        } else if (choice == 2) {
            remote = new AudioPlayerRemote();
            remote.switchOn();
            remote.play();
            remote.increaseVolume();
            remote.decreaseVolume();
            remote.switchOff();
        } else {
            System.out.println("Invalid choice");
            scanner.close();
            return;
        }
```

	Input	Expected	Got	
✔	1	TV Switched On TV Volume Increased TV Volume Decreased Tv Switched Off	TV Switched On TV Volume Increased TV Volume Decreased Tv Switched Off	✔
✔	2	Audio Player Switched On Audio Player Playing Audio Player Volume Increased Audio Player Volume Decreased Audio Player Switched Off	Audio Player Switched On Audio Player Playing Audio Player Volume Increased Audio Player Volume Decreased Audio Player Switched Off	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2

Correct

Mark 1.00 out of 1.00

Create an Interface A with methods work1() and work2(). Create a class AImplementation that implements the interface A. Now upgrade the interface A by adding one more method work3(), without affecting the class AImplementation.

For example:

Result
Work 1 Completed Work 2 Completed

Answer: (penalty regime: 0 %)

```
interface A{
    void work1();
    void work2();
    default void work3()
    {

    }
}
class AImplementation implements A
{
    @Override
    public void work1()
    {
        System.out.println("Work 1 Completed");
    }
    @Override
    public void work2()
    {
        System.out.println("Work 2 Completed");
    }
}
public class InterfaceUpgradeDemo
{
    public static void main(String [] args){
       AImplementation c =new AImplementation();
       c.work1();
       c.work2();
       c.work3();
    }
}
```

	Expected	Got	
✔	Work 1 Completed Work 2 Completed	Work 1 Completed Work 2 Completed	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.